



Inspiring Excellence

BRAC UNIVERSITY

CSE360: Computer Interfacing

Lab Project Report

Title: *Smart Emergency Response Assistant (SERA)*

by

Group: 3

Section: 1

Group Members:

Name	ID
SHAH MIRAN APURBO	22101799
RIYAJUL ISLAM	22101222
SHAHTI BINTI SORWARDI	21201235
NUSAIBA AHMED	22201883

Department of Computer Science and Engineering
BRAC University

Saturday, 06 September 2025

Abstract

The Smart Emergency Response Assistant (SERA) project aims to enhance the safety of homes by introducing a multi-purpose smart IoT-based solution to perceive and act on threats such as gas leaks, fire, earthquakes, and others. In this project, SERA will be linked to different types of sensors to detect any danger at any given moment, and these sensors include DHT11 (temperature/humidity), SW420 (vibration), MQ2 (gas), and a flame sensor, which includes one of the best microcontrollers (Arduino), which is cheap and effective. In the occurrence of dangerous conditions, an automatic emergency reaction will be triggered by the closing of the gas valve, opening the emergency door, switching on the audible alarm, and activating the LEDs. Alarms can be manually reset and acknowledged using a push button so that the user can control or clear the alarm state when necessary.

The modular method used by SERA reflects the successful implementation of computer-interfacing techniques to connect hardware and user interaction to provide more sophisticated safety mechanisms to smart homes and communities. Among the most important findings is reliable hazard detection, automation of physical response, and effective communication with users at relatively low cost. The sustainable design of the system encourages increased influence through the minimization of the risk of casualties, limited property damage, and allows a scalable and flexible response to a wide range of households. Finally, SERA promotes emergency preparedness and resilience and presents a sensible approach to next-generation smart home safety.

Keywords

Microcontrollers, IoT, Hazard Detection, Real-time Alerts, Computer Interfacing.

1. Introduction

Home safety risks such as gas leaks, fire breakouts, and earthquakes often occur with no prior notice which can result in significant loss of life and property. Many households lack the resources to implement integrated solutions for raising awareness of such incidents and responding effectively to them.

Our goal is to implement a Smart Emergency Response Assistant (SERA), an IoT-based system for hazard detection using sensors and real-time alerts through LEDs, alarms and messages shown on LCD screen.

The scope of this project is to implement computer interfacing techniques for integration of multiple sensors (gas, temperature, vibration) along with a microcontroller (Arduino) for data processing, decision making and communication. This project is envisioned as an implementation of computer interfacing between actuators and sensors and humans in real time that designs a practical safety system for domestic applications.

2. Related Work/ Inspiration

The inspiration of SERA was drawn from other previous systems such as commercial and amateur smart home systems that are able to identify hazards using sensors and feedback via IoT connectivity. According to the current systems, such as the safety platform developed by Smart Life Tech and other Arduino-based monitoring projects, the devices are typically optimized for reacting to only one kind

of hazard, such as fire, gas leakage, or intrusion. Many commercial programs integrate automated responses, such as activating sprinklers, opening doors, or shutting off electrical devices, to help control the disaster and assist people in safely evacuating the area.

Unlike most systems, SERA unifies several different hazard detection systems - gas leaks, fire and earthquakes - into one, single system using modest hardware, including the Arduino, common sensors and simple actuators. It is integrated with automatic response counter-measures (gas valve opening, door opening, fan ventilation), monitor of real time alert. Many related academic and commercial solutions rely on more expensive components or attempt to address only part of the hazards, are the local units only or lack automated physical intervention capabilities.

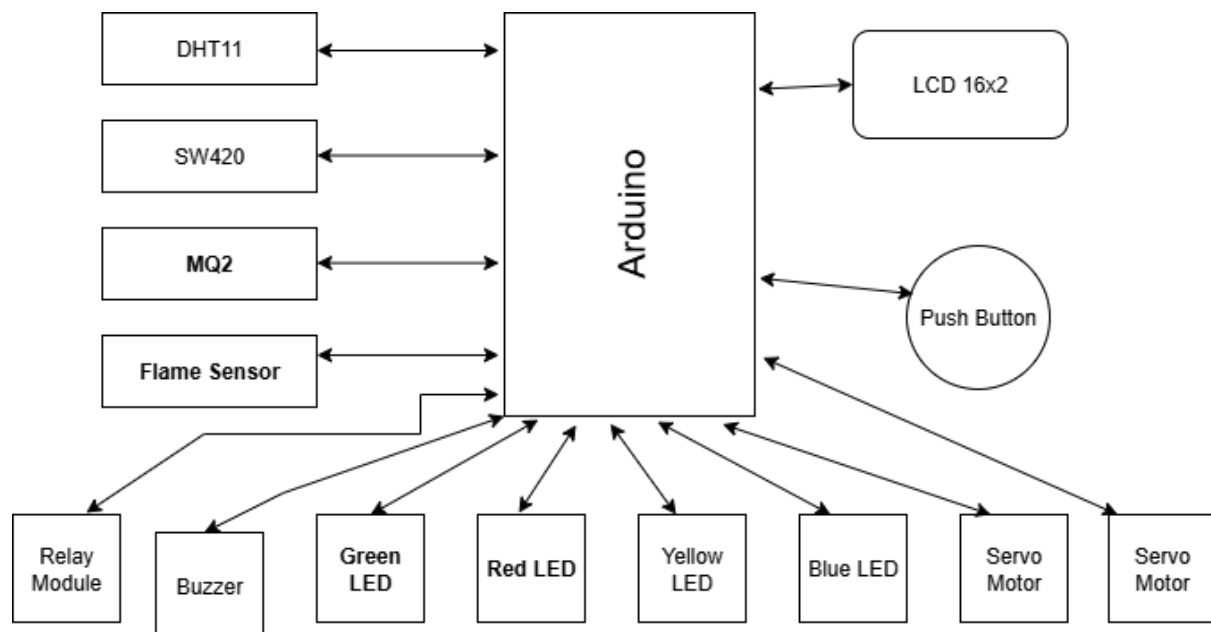
By contrast, SERA constitutes a low-cost and multi-hazard detection system that also ensures physical safety and notifications automatically in systems of low-cost buildings, for a group of people who are unable to have access to a high-tech safety system. This plan will address the failure of single hazard approaches, improve emergency preparedness, make existing safety solutions more accessible to vulnerable groups, and minimize false alarms and response time delay.

3. Technical Approach

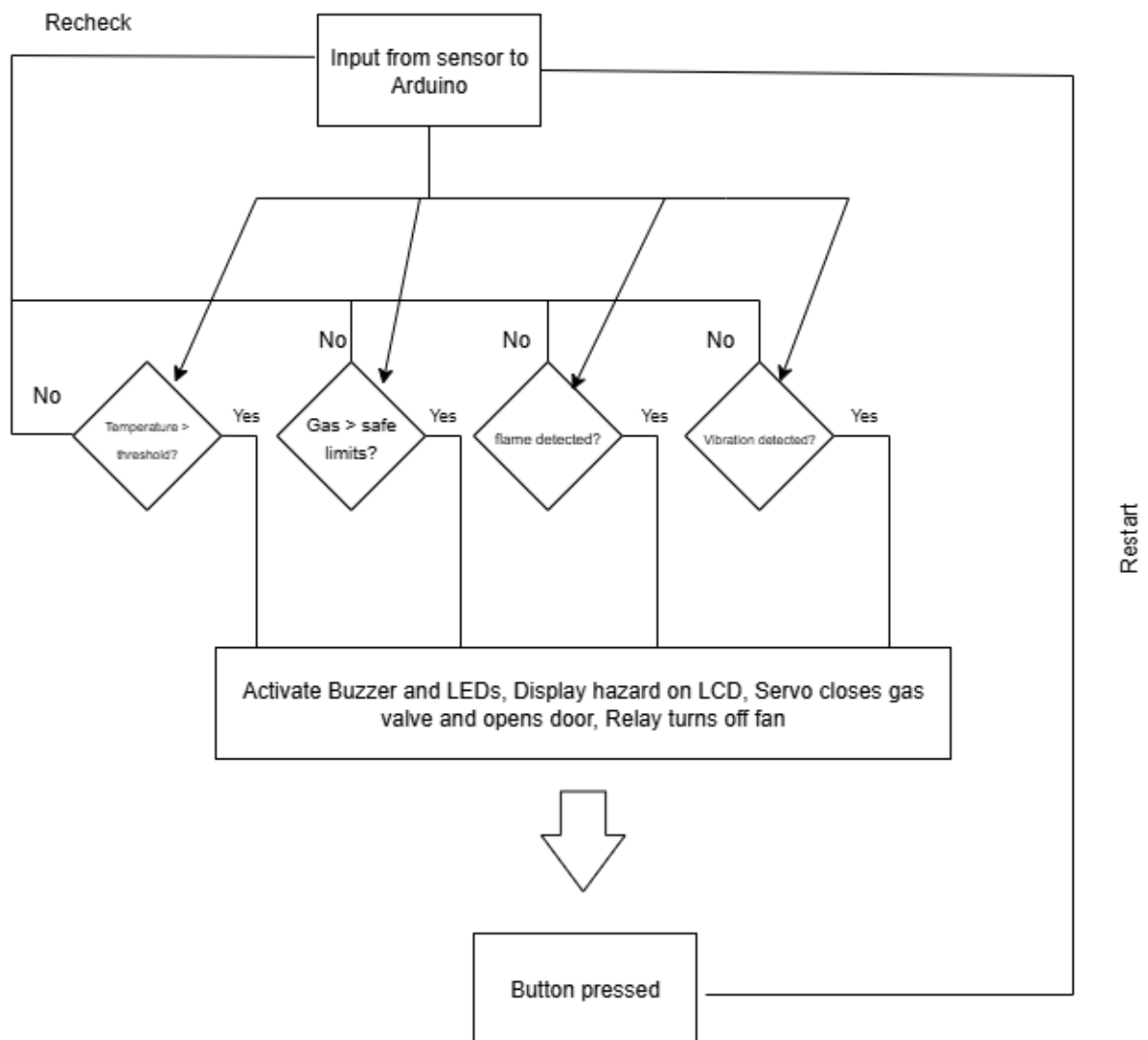
System Architecture:

SERA is an intelligent home safety system which uses IoT technology to sense and react to hazards in the home. Here, the Arduino takes data from the sensors of gas, humidity, temperature, fire, and vibration.

Block diagram:



Workflow:



Components Used:

1. DHT11 Temperature and Humidity Sensor
2. SW420 Vibration Sensor
3. MQ - 2 Gas Sensor
4. IR Flame Sensor
5. Dual Channel Relay Module
6. Buzzer
7. LEDs - Green, Red, Yellow, Blue
8. SG90 Servo Motor x2
9. 16x2 LCD Display with I2C module
10. Push Button
11. Resistors - 220, 100k

Cost Breakdown:

1. Arduino - Tk. 998
2. MQ2 gas sensor - Tk. 200
3. SW420 vibration sensor - Tk. 200
4. DHT11 - Tk. 150
5. Servo Motors (X2) - Tk. 450
6. Buzzer - Tk. 30
7. LEDs (X4) - Tk 23 (includes all Red, Green, Yellow, and Blue; each packet has 5 LEDs)
8. Flame Sensor - Tk. 65
9. Relay Module - Tk. 100
10. LCD 16X2 with I2C module - Tk. 340
11. Push button - Tk. 5
12. Jumper wires - Tk. 100
13. Breadboard - Tk. 145

Total = Tk. 2806

Communication Protocol:

In this system, a bunch of protocols are used to establish efficient communication among all the components.

1. I2C (Inter-Integrated Circuit):

I2C is a protocol that allows serial communication among multiple devices. It shares a common data bus with two wires only. One is SDA (Data) and another one is SCL (Shared System Clock). Thus, it reduces the wiring complexity with the Arduino.

We have chosen I2C for the LCD because it simplifies the wiring, resulting in the system communicating with the LCD using only two wires, reducing the number of GPIO pins needed on the Arduino.

2. Digital Communication (GPIO Pins):

The Flame sensor, SW-420 Vibration sensor, LEDs, and Buzzer use GPIO pins to send digital signals. This refers to the Digital Communication Protocol. All the data here is sent or received in a binary form with logical HIGH or LOW signals.

Most of our integrated sensors work on binary digital data. Besides, using the GPIO digital pins allowed faster detection due to raw binary outputs. We did not have to convert the raw signal. Also, this protocol was the simplest and most suitable for our objectives.

3. Analog Communication (ADC):

Analog Communications also uses an ADC to digitize the analog signal to make it computer readable by the controller. This standard is important whenever we would like an accurate, continuous value output from a sensor.

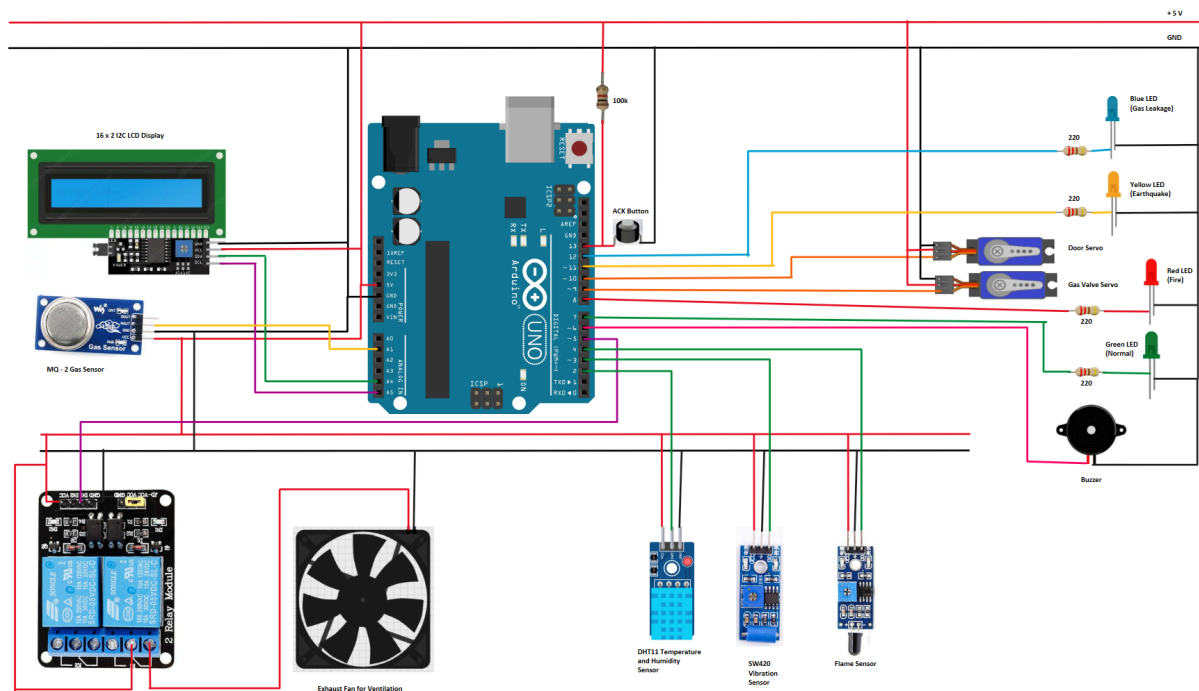
Here MQ-2 sensor provides a raw continuous value from the environment which should be converted to a digital signal. That is why this protocol has been selected for the MQ-2 sensor to communicate with controller.

4. Pulse Width Modulation (PWM):

Pulse Width Modulation (PWM) is used for precise control of any actuator.

In our system, we used SG90 servo motors to manipulate the position of the gas valve and emergency exit in the prototyped room. So we needed precise movement similar to a discrete format. So we choose a PWM frequency to send the controlling signals to the servo motors.

Circuit/Schematic:



Implementation:

1. Arduino is used to initialize the system, which includes setting up the sensors and relay to begin collecting data.
2. The Sensors DHT11, SW420, MQ2, Flame Sensor are constantly monitoring for abnormalities in their surroundings.
3. The output from the sensor is then compared to threshold values to see if there is any output above the threshold.
4. If there is data detected over the threshold, LEDs and buzzers are turned on, servo motors turned on for closing gas valves and opening doors, and the relay fan is powered off.
5. The LCD communicates status of the hazard to the user. This can be noted by the user at the button to reset the system.

Challenges:

Challenge	Explanation	Implemented Solution
False alarms from single sensors	Sometimes individual sensors (such as MQ-2 or DHT11) can cause false readings which aren't necessary to generate unwanted hazards.	Combines multiple sensors sensibly before taking actions Only as the fire is detected by the high temperature and flame sensor, it activates the fire alarm.
Unresponsive acknowledgement button	In the old handlers button detection was blocked by very long delay() calls in the hazard handling.	Added checkAcknowledgementButton () in several places and made sure they are checked after handling a hazard. This means that a manual reset is possible right away.
LCD flickering and slow updates	lcd.clear() in each iteration of the loop, did flickering.	Limited lcd.clear() calls only inside hazard handling functions. Normal updates use lcd.setCursor() without clearing the screen.
Single-point failure risk	The system could become unsafe if one actuator or sensor malfunctions.	Layering of sensors - each hazard has several sensors, and actuators are isolated from each other (e.g. fan, gas valve, door).
Multiple hazards occurring simultaneously	Fire, gas leak and earthquake might occur at the same time.	Added various functionalities (handleAll(), handleFireGasLeak(), etc.) to include combined hazard conditions.
Timing of hazard response	Actions must last long enough that users will be aware of them; otherwise, actions that are performed too quickly may not even be noticed.	Added delay() to hazard handling functions (can be optimized later with millis() which is non-blocking response).

4. Sustainability & Impact

Sustainability: SERA has been programmed with Arduino and is characterized by low energy usage and power consumption. The items and the sensors are small and long lasting in the system leading to an overall savings of energy in the home. The casing and mounting of the material is carried out where practicable with recyclable and reusable materials to minimize waste and impact on the environment. In order to reduce the electromagnetic pollution that is a hallmark of the IoT systems, SERA facilitates wired communication choices and limits the emissions of wireless communications.

Impact: SERA could be a great positive influence on society. It assists in identifying various risks in a more cost effective way. In case of disaster such as earthquakes, fire and gas leakages in hazardous areas, SERA can be helpful in responding to automated emergency systems. It will result in reduced destruction and increased lives saved. It can have a significant impact on residential, industrial, and public space due to its service-oriented design.

Future Work:

- Increased scalability and the ability to control the device remotely by use of wireless communication modules.
- Addition of sensor types: smoke, flood, or intrusion detection.
- AI-based predictive hazard detection and intelligent automation analytics.
- Adding more notifications, control in a better user interface, and additions of mobile apps.
- Cloud connection, which can be centrally managed and deployed on a large scale in smart cities or in corporate complexes.

Limitations:

- Hardware constraints, such as limited processing power and memory of the Arduino, can slow response time or reduce the number of devices that can be connected to it.
- Applied sensors may be simple sensors and may not be able to detect all hazardous conditions, or may lack sensitivity.
- Software constraints include simple logic and an inability to make advanced decisions, and even an inability to operate with higher standard smart home protocols.
- No backup power systems, so the system cannot be used when there is an outage.

5. Results & Discussion

Simulation: We successfully managed to simulate scenarios to see if SERA would give the desired results based on our logic and circuit design. The following scenarios were made up to test for accuracy: Normal scenario, Gas leak scenario, Fire scenario, and Earthquake scenario.

1. **Normal Scenario:** The temperature was kept around 28-29 degrees (room temperature), and the humidity level was also kept at a normal level of 55%. No flames were kept near the system, and no vibration was simulated (by shaking the table).

System Response: Green LED on, which means nothing unusual was detected, LCD screen displayed 'System Normal' with temperature and humidity readings, and no alarms were triggered.

2. **Gas Leak Scenario:** Gas levels were raised to 650 (above threshold)

System Response: Blue LED was on, buzzer activated, notifying abnormal gas levels, LCD displayed "Gas Leak Detected." The servo closed the gas valve, and the exhaust fan turned ON for ventilation.

3. **Fire Scenario:** A small fire from a lighter was placed near the system, temperature was recorded to be around 45 degrees or more.

System Response: Red LED on, detecting fire, buzzer activated, LCD displayed “Fire Detected.” The servo motor opened the door, relay switched the fan ON.

4. **Earthquake Scenario:** The table was shaken, and SW420 was able to detect vibration, potentially detecting earthquakes.

System response: Yellow LED on, buzzer activated, LCD displayed “Earthquake Detected.” The servo motor opened the door automatically for the emergency exit.

Sample Test Data:

Condition	Temp (C)	Humidity (%)	Gas Level	Flame	Vibration	System Response
Normal	29	55	180	No	No	Green LED ON, LCD: Normal
Gas Leak	32	58	650	No	No	Red LED, buzzer, valve closed, notification sent
Fire	45	53	300	Yes	No	Red LED, buzzer, fan ON, door opened, notification sent
Earthquake	30	56	190	No	Yes	Blue LED, buzzer, door opened, notification sent

Sensor Input and Output Truth Table:

Temperature > 30	Gas level > 500	Vibration Detected	Flame Detected' (Active LOW)	Output
0	0	0	0	Normal
0	0	0	1	Normal
0	0	1	0	Earthquake only
0	0	1	1	Earthquake only
0	1	0	0	Gas leak only, door open
0	1	0	1	Gas leak only, door open
0	1	1	0	Gas leak and Earthquake
0	1	1	1	Gas leak and Earthquake
1	0	0	0	Fire
1	0	0	1	Fire detected; Ventilation
1	0	1	0	Fire and Earthquake
1	0	1	1	Ventilation and Earthquake detected
1	1	0	0	Fire and Gas leak
1	1	0	1	Gas Leak and Ventilation
1	1	1	0	Fire, Gas leak, and Earthquake
1	1	1	1	Fire, Gas leak, and Earthquake

6. Conclusion

Low-cost integrated equipment such as SERA is designed to detect and respond to domestic hazards in time. To improve user awareness, the project has been capable of demonstrating reliable multi-hazard monitoring, physical security, including door and gas valve control, and real-time remote notification, using a wide variety of sensors and smart automation. These positives serve to bridge the

divide between family and community members who do not have access to more advanced safety systems and help to increase general preparedness on emergency and response resources.

Not only is the impact on the environment limited by the sustainable design of SERA and the use of energy-saving and recyclable materials, but the design also increases the possibilities for the layers of the population most susceptible to the impact of numerous people. It has a positive footprint in the societal impact perspective, as it reduces vulnerabilities on the household level, provides timely evacuation and response, and extends to the village or industrial level with limited resources. And finally, we can also sell the ethics of the modern smart house by way of SERA: safety, sustainability, and a smart house's ability to include everyone.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
SHAH MIRAN APURBO	22101799	Hardware design, Circuit building, and testing	- Installed and prepared the physical connections of all the sensors (DHT11, MQ-2, SW-420, Flame sensor) to Arduino UNO. - Wired and programmed the actuators (servo motors, exhaust fan relay, LEDs and buzzer) to the Arduino board. - Proper distribution of power among components to prevent overloading of Arduino or other devices attached to it (sensors, actuators, etc.). - Individual components and wiring was tested and verified to not have any connection problems. Assure the sensors provide us with valid data, and actuators are responding to simple commands. - Confirmed that the power supply was sufficient to power all the components, especially the more power-intensive components such as the servo motors and relay modules.
RIYAJUL ISLAM	22101222	Software Design	- The system was written in the IDE with the entire CPP code. - Created the code to read the information of all sensors (DHT11, MQ-2, Flame sensor, SW-420). - Introduced the logic to synthesize sensor data and

			make decisions regarding how the system should react to different hazardous situations (fire, gas leak, earthquake). • Check all sensor reading functions, hazard response code, debug, timing problems, bad sensors, etc., and verify that the system responds correctly to hazards.
SHAHTI BINTI SORWARDI	21201235	Prototype Design and Communication Protocol Integration	• Choosing the right sensors and actuators depending upon the project needs, keeping in mind the cost, dependability, and integration. • Select a communication protocol most suitable to devices. I2C is an example, LCD display, digital/analog sensors, and actuators. • Connecting the sensors to the Arduino (using their corresponding protocol (analog/digital/I2C)) and the actuators to the Arduino (via the PWM (servos) or digital output (the relay and the LEDs))
NUSAIBA AHMED	22201883	Prototype Design and Actuator Integration	• Install the relay unit with the exhaust fan and other high-power output units so that when the hazard is detected, the fan is switched by the relay. • Provided that the actuators (servo, relay, fan) are all fault-tolerantly controlled to prevent a system failure, for example: the gas valve cannot open in the event of a fire. • Tested each actuator physically to make sure that it is responding to the Arduino commands correctly. As an example, test the servos and make sure they close and open the door and gas valve properly.

References

- United Nations Office for Disaster Risk Reduction. (2023, June 6). Words into Action: Multi-hazard early warning systems [Video]. YouTube.
<https://www.youtube.com/watch?v=vEgukHtd79o>
- Rao, S. T. L., Jyoshna, K., Lokesh, A., Sruthi, D., & Jeevansai, N. (2025). Smart environment and safety monitoring system using Arduino. International Journal of Engineering Research & Technology (IJERT), 14(5).
<https://www.ijert.org/smart-environment-and-safety-monitoring-system-using-arduino>
- International Journal of Scientific Research in Computer Science, Engineering and Information Technology. (2024). Early detection system for gas leakage and fire in a smart home using IoT. Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol., 10(3), 102–108.
<https://doi.org/10.32628/CSEIT24102123>
- Quectel Wireless Solutions. (2024, March 21). Mass scale deployments unleash the socio-economic benefits of IoT.
<https://www.quectel.com/blog/mass-scale-deployments-unleash-the-socio-economic-benefits-of-iot/>